20. The reagent used to distinguish between acetaldehyde and benzaldehyde is

a) Tollens reagent b) Fehling's solution

c) 2,4 – dinitrophenyl hydrazine d) semicarbazide

21. Phenyl methanal is reacted with concentrated NaOH to give two products X and Y. X reacts with metallic sodium to liberate hydrogen X and Y are

a) sodiumbenzoate and phenol b) Sodium benzoate and phenyl methanol

c) phenyl methanol and sodium benzoate d) none of these

22. In which of the following reactions new carbon – carbon bond is not formed?

a) Aldol condensation b) Friedel craft reaction

c) Kolbe's reaction d) Wolf kishner reduction

23. An alkene "A" on reaction with O_3 and Zn - H_2O gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is

a) $Cl-CH_2-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{CH}}}}$

b) $H_3C-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_2Cl}{\overset{|}{\underset{|}{CH}}}}$

c) $H_3C-CH_2-\underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_3$

d) $H_3C-\underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{CH}}}}-CH$

24. Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their (NEET)

a) more extensive association of carboxylic acid via van der Waals force of attraction

b) formation of carboxylate ion

c) formation of intramolecular H-bonding

d) formation of intermolecular H – bonding

Short Answer Questions

1. How is propanoic acid is prepared starting from

(a) an alcohol (b) an alkylhalide (c) an alkene

2. A Compound (A) with molecular formula C_2H_3N on acid hydrolysis gives(B) which reacts with thionylchloride to give compound(C). Benzene reacts with compound (C) in presence of anhydrous $AlCl_3$ to give compound(D). Compound (D) on reduction with Zn/Hg and Conc.HCl gives (E). Identify (A), (B), (C), (D) and (E). Write the equations.

3. Identify X and Y.

$$CH_3COCH_2CH_2COOC_2H_5 \xrightarrow{CH_3MgBr} X \xrightarrow{H_3O^+} Y$$

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